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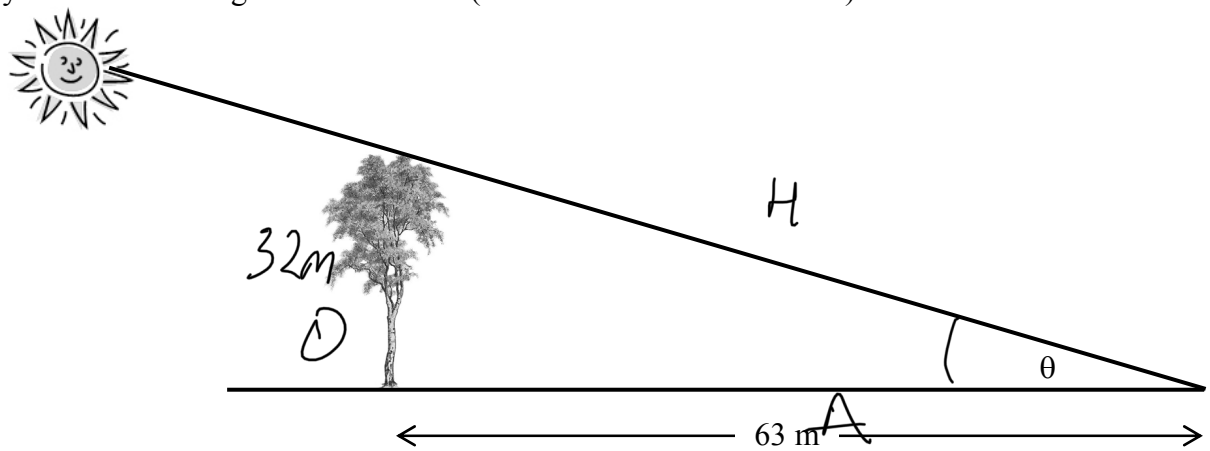
**Question 13** **(Suggested maximum time: 5 minutes)**

A tree 32 m high casts a shadow 63 m long. Calculate θ, the angle of elevation of the sun.
Give your answer in degrees and minutes (correct to the nearest minute).




SOCAHTOA

$$\tan\theta = \frac{32}{63}$$

$$\theta = \frac{\frac{32}{63}}{\tan} = \tan^{-1}\left(\frac{32}{63}\right) = 27^\circ$$

$$3x = 9$$

$$x = \frac{9}{3}$$

$$4^{-1} = \frac{1}{4}$$